Assignment 2

Mark: 20

Deadline: 3 December 2025 Wednesday 11:00 PM

Submission link: https://forms.gle/6JbitXpuwnRmk7P18

Solve two problems (one from each category)

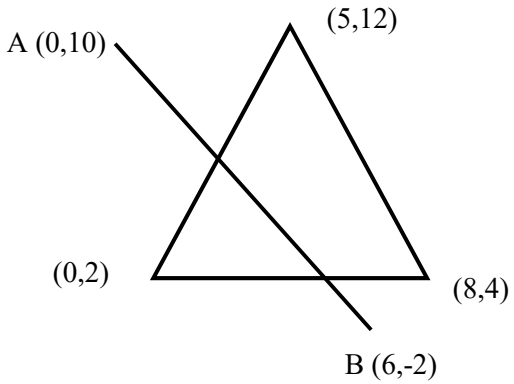
Category A: (solve one)

1. Given (-30, -30) and (30, 30) are the corners of the clip rectangle. The endpoints of a line are given as (35, 37) and (-38, -20). Using the Cohen-Sutherland line clipping algorithm, identify whether the lines are Accepted, Rejected, or Partial (partially accepted) only from the outcodes. Does the Cyrus-Beck line clipping algorithm produce the same output?

2. Given (0, 0) and (400, 300) are the corners of the clip rectangle. The endpoints of a line are given as (−50, 350) and (450, 330). Using the Cohen-Sutherland line clipping algorithm, identify whether the lines are Accepted, Rejected, or Partial (partially accepted) only from the outcodes. Does the Cyrus-Beck line clipping algorithm produce the same output?

3. The top-right corner of a clipping rectangle is (20, 60). The window is 60 units wide and 40 units tall. The endpoints of a line segment are (7, 16) and (85, 75). Apply the Cohen-Sutherland Algorithm, and identify whether the line is "partially inside", "completely inside", or "completely outside". If it is the first case, run the algorithm to calculate new endpoints for the line segment so that it is inside the clipping window. Does the Cyrus-Beck line clipping algorithm produce the same output?

4. Using Cyrus Beck algorithm, find out t values of the line AB and specify whether they are entering or leaving using Cyrus Beck algorithm. Also find the line segment within the clip triangle.



Category B: (solve one)

1. When the reflected with respect to the line $y = 3x - 8$, the coordinate of the point becomes (3,5). What is the original coordinate of the point?
2. A 3D point M is transformed to M' applying shearing along the Y Axis by a shearing factor of (12, 21) with respect to a point (-33, 44, -55). If the **new coordinate M'** is (125, 255, -674). What was the **original coordinate** of **M**?
3. A robot is positioned at (2, 0, 3) with the endpoint of its hand at (2, -2, 1). The robot wants to grab a fruit located at (12, 5, 13). To achieve this, it performs the following transformations ***sequentially*** on its hand's endpoint:
   - **Rotation:** At first, the hand's endpoint is rotated 90° counter-clockwise around the X-axis, using the robot's position as the center of rotation.
   - **Scaling:** Then, the hand's endpoint is scaled by some factors of **S_x, S_y, and S_z.**
   - **Translation:** Lastly, the robot itself is translated by (10, 0, 15), bringing the hand's endpoint to the fruit's position so that it can grab the fruit.

   Find the **values of S** that ensure the hand's endpoint aligns with the fruit (the hand's endpoint will be the same as the fruit's location). Show your calculations using the composition of transformations. Also, explain whether the scaling is uniform or non-uniform.
4. Derive |4x4| rotation matrix for a 3D point while;
   a. Rotation across the X-axis and the center of rotation (a, b, c).
   b. Rotation across the Y-axis and the center of rotation (a, b, c).
   c. Rotation across the Z-axis and the center of rotation (a, b, c).

Number distribution:

Within deadline	Full marks
1 day late	-2
2 day late	-4
>2 day late	Won't be accepted